ESP32 XIAO Wi-Fi Flight Controller for Micro Drone

Project Description

The **ESP32 XIAO Wi-Fi Flight Controller** is a compact, custom-designed flight control system developed for lightweight micro drones powered by **6×15 mm coreless brushed motors**. Built around the **Seeed Studio ESP32 XIAO**, the controller combines wireless communication, flight stabilization, and motor control into a small, efficient PCB suitable for educational, research, and hobbyist UAV applications.

The flight controller integrates an **MPU6050 6-axis Inertial Measurement Unit (IMU)** via the **I^2C interface** to provide real-time motion sensing for attitude estimation and flight stabilization. Four custom MOSFET-based motor driver circuits independently control the brushed motors, while onboard status LEDs provide visual feedback during operation.

The system supports two flight modes. The first uses **Wi-Fi communication** through the ESP32, implementing the **Crazyflie UTP protocol**, allowing the drone to be controlled wirelessly using a computer or mobile device. The second supports **BetaFlight firmware**, enabling advanced flight tuning, configuration, and FPV-style control similar to commercial racing drones.

Designed as an open-source hardware platform, the project provides complete PCB schematics, firmware resources, and documentation, making it an excellent platform for learning embedded systems, flight control algorithms, PCB design, and drone development.

Hardware Components

Component	Quantity	Purpose
Seeed Studio ESP32 XIAO	1	Main microcontroller with Wi-Fi connectivity
MPU6050 IMU	1	3-axis gyroscope and 3-axis accelerometer for flight stabilization
SI2300 MOSFET	4	Drives the brushed motors
1N4148WS Diode	4	Motor driver protection and switching
Coreless Brushed Motors (6×15 mm)	4	Drone propulsion
LEDs (Red, Green, Blue)	Multiple	Status indication and visibility
10 kΩ Resistors	Multiple	Pull-up and biasing circuits
100 Ω / 150 Ω / 200 Ω Resistors	Multiple	Current limiting and signal conditioning
10 nF Capacitors	Multiple	Noise filtering

Component	Quantity	Purpose
100 nF Capacitors	Multiple	Power supply decoupling
Custom PCB	1	Flight controller hardware integration
Li-Po Battery	1	Portable power supply

Key Features

- Compact ESP32 XIAO-based flight controller
- Integrated MPU6050 IMU for attitude sensing
- Four custom MOSFET motor driver circuits
- Supports four 6×15 mm brushed motors
- Wi-Fi-based drone control using the Crazyflie UTP protocol
- Compatible with BetaFlight firmware and configurator
- Low-latency wireless communication
- Open-source PCB design and firmware
- Expandable for FPV cameras and external radio receivers
- Lightweight design optimized for micro drones

Applications

- Micro quadcopter development
- Educational UAV platforms
- Flight controller prototyping
- Embedded systems and PCB design learning
- Wi-Fi-controlled drones
- FPV racing drone experimentation
- Robotics and autonomous flight research
- Open-source drone development